FEDERAL SPECIFICATION

BOTTLE, STOPPER, MILK DILUTION, 160 MILLILITERS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a 160 milliliters (ml.) milk dilution bottle, with Escher type rubber stopper, suitable for use in the preparation of milk dilutions prior to plating.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.**—The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-585 — Boxes, Wood, Wirebound.
PPP-B-601 — Boxes, Wood, Cleated-Plywood.
PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 — Boxes, Fiber.
PPP-B-676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102 — Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S Government Printing Office, Washington 25, D. C

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies)

Military Specification:

MIL-L-10547 — Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Public Health Association:

Standard Methods for the Examination of Dairy Products.

(Copies may be obtained upon application to the American Public Health Association, 1790 Broadway, New York 19, N. Y.)

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American Society for Testing Materials:
ASTM C148–50 Standard Method of Polariscopic Examination of Glass Containers.
ASTM C225–54 Standard Methods of Test for Resistance of Glass Containers to Chemical Attack.

(Applications for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Glass.*—Bottle shall be made of good quality, clear, borosilicate glass which shall be capable of withstanding successfully all tests hereinafter specified.

3.1.2 *Rubber stopper.*—Stopper shall be made of a soft, black natural rubber compound, and shall be free of cuts, tears, ruptures, or other defects.

3.2 Style and design.—The bottle shall be square type, complete with Escher rubber stopper, specially designed for the dilution of milk samples prior to plating, as specified in the latest edition of "Standard Methods For The Examination of Dairy Products", of the American Public Health Association, Inc. The bottle shall be designed with a heavy circular ridge around the outside of the bottle bottom, and shall be stable on a flat surface.

3.3 Dimensions. — Dimensions of bottle shall be as shown on figure 1. The body wall shall be a minimum of 0.062 inch and the bottom a minimum of 0.078 inch.

3.4 Capacity and graduation.—The bottle shall have a capacity of 160 ml., and shall be permanently marked on one side with a fine engraving wheel or abrasive blasting (unfilled) or with a fine etched enamel filled line of not more than 0.5 millimeters (mm.) thick and so positioned as to indicate a capacity of 99 ml. ± 1 ml. when tested as specified in 4.4.5 (see figure 1). Enameled graduation line shall show no change, or removal, when subjected to the tests specified in 4.4.3.

3.5 Area for marking.—There shall be a circular area on one side, roughened by sandblasting, etching, suitably treated or spot enameled to provide for marking. The area, ¾ inch ± 1/16 inch in diameter, shall be positioned as shown in figure 1. The marking area shall show no change when subjected to the tests specified in 4.4.3.

3.6 Resistance to thermal shock.—Upon removal from the autoclave or from the cold immersion bath, bottles shall not chip, crack, or show evidence of clouding when tested as specified in 4.4.1.

3.7 Solubility.—The solubility of the glass shall be such as to require not more than 1.0 ml. of 0.020*N* sulfuric acid to neutralize the extract from 10 grams of glass when tested as specified in 4.4.2.

3.8 Annealing.—Bottles shall be well annealed when tested as specified in 4.4.6. When examined with a polariscope, there shall be no ring strain or severe longitudinal strain. Any longitudinal strain present shall be faint and highly diffused.

3.9 Rubber stopper. — The milk dilution bottle shall be furnished with a rubber stopper of the Escher type in conformance with figure 2. The stopper shall be size 2 and shall have an extension of 42 mm. ± 1 mm. in length which shall fit down into the bottle to prevent its displacement during sterilization. The rubber stopper shall not blister, bleed, bloom or show any deformation when subjected to autoclaving as specified in 4.4.4.

3.10 Identification marking.—Bottle and stopper shall be permanently and legibly marked with name or registered trademark of the manufacturer.

3.11 Workmanship. — Workmanship shall be first class throughout. Glass shall be uniformly distributed, and free from such defects as bubbles, seeds, blisters, striae or other irregularities which detract from its

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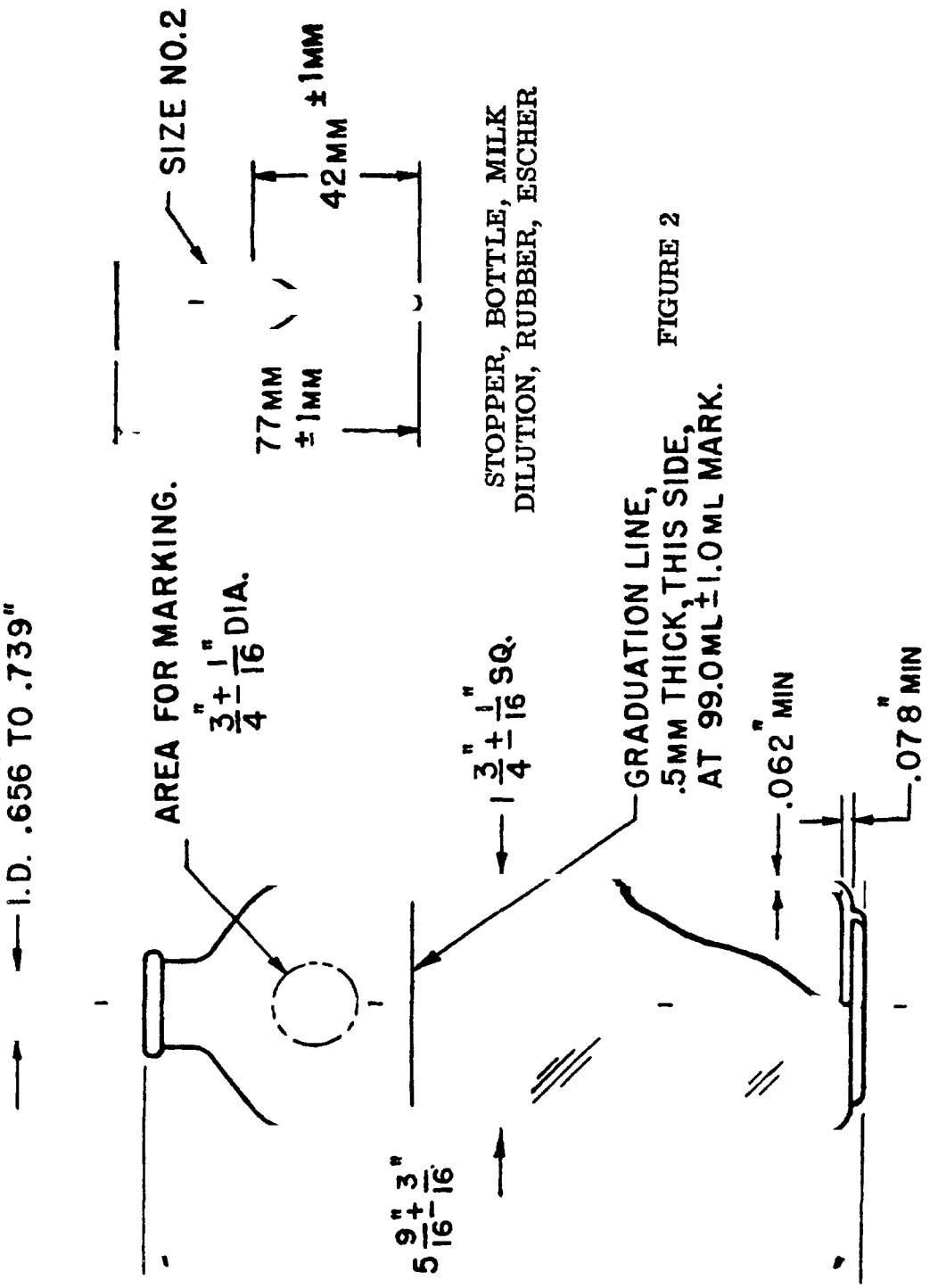


FIGURE 2

BOTTLE, STOPPER, MILK DILUTION, 160 MILLILITERS
FIGURE 1

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appearance or may impair its use or serviceability. The bottles at time of shipment, shall be clean, dry, and ready for use.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. —Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as, visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.*—When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. — Sampling shall be conducted in accordance with Military Standard MIL–STD–105 and Appendix thereto.

	Inspection level	AQL (percent defective)
For visual examination:		
Major defects	II	2 5
Total (major and minor combined)	II	6 5
For dimensional examination	L3	4 0

For end item testing:

Characteristic	Requirements	Test procedure	Inspection level	AQL
Capacity at graduation mark	3.4	4.4.5	L3	1.5
Permanency of marking area	3.5	4.4.3		
Alkali test:				
Sodium hydroxide		4.4 3.1	L1	6.5
Acid tests:				
Hydrochloric		4 4.3.2	L1	6 5
Sulphuric		4 4.3 3	L1	6.5
Thermal shock test	3.6	4 4.1	L3	1.0
Solubility test	3.7	4.4 2	L1 Min samples = 2	*
Annealing-ring and severe longitudinal strain	3.8	4.4.6	L3	1 0
Stopper autoclaving	3.9	4.4.4	L3	2.5

* Each of the bottles sampled shall be submitted to its own individual test. Acceptance number shall be zero for all sample sizes.

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4.2.1 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 **Examination.**—The bottles with stopper shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 *Classification of visual defects.* — Examination shall be conducted in accordance with the following classifications of defects:

Categories and defects

Major

101. Glass not free of blisters, bubbles, cracks or deformed areas.
102. Bottle not stable on flat surface.
103. Graduation etching not present
104. Area for marking not present
105 Area for marking not as specified.
106. Heavy circular ridge base not on exterior bottom of bottle.
107. Rubber stopper not free of cuts, tears, ruptures or other defects.

Minor

201. Bottle not clean and dry.
202. Glass not clear.
203 Bottle not free from indentations, wrinkles, tears, seeds, stones or dirt.
204. Permanent identification marking not present on bottle and stopper.
205. Bottle walls free of localized thin areas.
206 Bottle not free of rough surfaces.
207. Stopper not black.

4.3.2 *Dimensional examination.*—The bottles and stoppers shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 **Tests of end item.**

4.4.1 *Thermal shock test.*—The bottles shall be heated in an autoclave for 15 minutes at a temperature of 121° centigrade (C.) and 15 pounds per square inch (p.s.i.). When removed from the autoclave, the bottles shall be immediately immersed in water of 3° to 5° C.

4.4.2 *Solubility test.*—The solubility shall be determined in accordance with ASTM C225–54 (method P–W).

4.4.3 *Permanency of area marking.* — In performing the following tests, a different bottle shall be used for each test:

4.4.3.1 *Alkali test.*—Immerse sample completely in normal sodium hydroxide at room temperature, cover, and let stand 1½ hours. Remove sample and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.4.3.2 *Acid test (hydrochloric).*—Immerse sample completely in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, and let stand for 1½ hours. Remove sample and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.4.3.3 *Acid test (sulphuric).* — Immerse sample completely in concentrated sulphuric acid at room temperature and let stand 1 hour. Remove sample and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

4.4.4 *Resistance to autoclaving.*—Rubber stoppers shall be subjected to autoclaving for 30 minutes at a temperature of 121° C. and 15 p.s.i.

4.4.5 *Test for capacity to graduation mark.* —A 100-ml. burette shall be filled to indicated capacity with distilled water at 20° – 25° C. and allowed to flow into the sample bottle until the bottom of the meniscus is level with the top of the graduation mark. The total volume of distilled water to the graduation mark of the bottle shall be 99 ml. ± 1 ml.

4.4.6 *Annealing test.*—Annealing shall be determined in accordance with ASTM No. C148–50.

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5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*

5.1.1.1 *Intermediate package.*

5.1.1.1.1 *Bottle.*—Twelve (12) bottles shall be intermediately packaged in a fiberboard box of appropriate size and design constructed in accordance with Federal Specification PPP–B–636, type I, class 1 or 2. The container shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.1.1.2 *Stopper.*—Twelve (12) stoppers shall be intermediately packaged in a carton of appropriate size and design constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.2 *Level C.*—Bottles and stoppers shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of these products for their intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.* — Twelve (12) containers of bottles or forty-eight (48) containers of stoppers, packaged as specified in 5.1.1.1.1 and 5.1.1.1.2, shall be packed in a wirebound wood, cleated-plywood or nailed wood box conforming to Federal Specifications PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; and PPP–B–621, class 2, respectively. Boxes for bottles shall be designed for type 3 load and boxes for stoppers designed for type 2 load. Each box exterior shipping container shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing of the liner shall conform to applicable paragraphs of the appendix thereto. Strapping shall be in accordance with the applicable box specification.

Note—Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*—Twelve (12) containers of bottles or forty-eight (48) containers of stoppers, packaged as specified in 5.1.1.1.1 and 5.1.1.1.2, shall be packed in a nailed wood box conforming to Federal Specification PPP–B–621, table I, or a fiberboard box conforming to Federal Specification PPP–B–636, type I, class 1, constructed of not less than 275-pound test fiberboard.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2.4 *Packing variation permitted.*—If the required number of bottles and/or stoppers to be shipped is less than the number of bottles and/or stoppers specified to be overpacked in an exterior container, such bottles and/or stoppers shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking required in the contract or

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order, each intermediate and shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*

5.3.2.1 *Intermediate packages.*—Each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2.2 *Exterior container.*—Exterior container shall be marked in accordance with Military Standard MIL–STD–129, except that 6.2(c), 6.2(d), 6.5, and 7.5.1 shall not be applicable. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items appearing in the Armed Services Medical Stock List:

8125–404–8600 BOTTLE, STOPPER, Milk Dilution, 160 ml.

8125–404–8650 STOPPER, BOTTLE, Milk Dilution, Rubber, Escher.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.4 Transportation description.—Transportation description applicable to this item is:

Bottles, not otherwise indexed by name, glass, other than cut, one quart or less in capacity.

Carload minimum weight 30,000 pounds.

Bottles, not otherwise indexed, glass, other than cut, one quart or less in capacity.

Motor volume minimum weight 30,000 pounds.

6.5 Supersession data.—This specification includes the requirements of Military Specification MIL–B–3326 dated October 31, 1950.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each

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